

Computer Graphics I, Project #1, A Drawing Panel

1. Design an interactive program allowing users to draw 2D geometries and type texts. The program should have a graphical user interface (GUI) via which users can define attributes, select geometry types, chose fonts, and draw objects.
2. 說明:The program opens a window in which users draw 2D objects with different **attributes**. The objects may be:
 - a 、 Polygons, circles, lines, points, texts and other 2D objects.
3. Basic requirements (70%)
 - (a). Include **menus** in your program. (Hierarchical menus)
 - (b). Handle *keyboard*, *mouse*, *window* and *motion* events in your program.
 - (c). By using the menu system, users can change colors, line width, point size, fill-mode, etc.
 - (d). By using motion and mouse events, users can control the drawing, for example, defining vertex positions for polygons.
3. Advanced requirements: (40%)
 - (a) Save the contents of color buffer by using *idle* events (or timer).
 - (b) Copy the saved color buffer contents while a display event occurs.
 - (c) Allow users to save the color buffer. (15%)
 - (d) Allow users to type in strings. (5%)
 - (e) Draw grid-lines or scales to show positions and sizes of objects. (5%)
 - (f) Other fancy ideas are welcome. (15%)
4. You can modify our sample programs to create your drawing program.
5. Please write the manual of your program in A4 size papers and hand in the papers when demo your program. (要訂好，含心得、操作方法、特殊貢獻與其他有利於評分的資訊)
6. You have to demo your system in the Graphics Lab. Two weeks later.
7. No cheating! 助教會在 Demo 時，詢問程式內容與設計方法。
8. 利用其他 *libraries* 完成的功能，不加分，也不算分。